

UNPHASED

Version 3.1.7

User guide

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Introduction

UNPHASED is an application for performing genetic association analysis in nuclear families and unrelated subjects. It implements maximum-likelihood inference on haplotype and genotype effects while allowing for uncertain phase and missing genotypes. Many of the commonly performed analyses are provided including transmission/disequilibrium tests, global and individual tests for haplotypes, tests that account for associations of nearby loci, tests of gene-gene interaction, adjustments for environmental covariates, genotype tests, comparison of risk between different haplotypes, and permutation tests.

If you use UNPHASED for a publication, please acknowledge my work. The main reference for UNPHASED is:

Dudbridge F (2008) Likelihood-based association analysis for nuclear families and unrelated subjects with missing genotype data. *Human Heredity* 66:87-98

The reference for the –sibship option is:

Dudbridge F, Holmans PA, Wilson SG (2011) A Flexible Model for Association Analysis in Sibships with Missing Genotype Data. *Ann Hum Genet* 75:428-38

UNPHASED is distributed free-of-charge under the GNU public licence. The latest version of UNPHASED is available from

<http://sites.google.com/site/fdudbridge/software/>

Running UNPHASED

Installation

After downloading the distribution, extract the files to an appropriate directory. On Windows systems, use the **Extract All** facility for zipfiles. On unix systems, type

unzip unphased-3.1.7.zip

Five directories are created:

1. **bin/** contains executable files for Linux, Windows and Mac machines, and a Java bytecode for the graphical user interface.
2. **doc/** contains this documentation and two open access papers describing the theory.
3. **src/** contains the source code, from which you can compile your own executable.

Binaries are provided for your convenience but use them at your own peril. You may find that the program runs faster if you build it on your own system – the easiest way to do so is with a **make** utility. The distributed executables were built using g++, under mingw for Windows and Xcode for Mac. The graphical user interface should work on Java Virtual Machine 1.5 or later.

If you are using Linux or Mac machines, you may want to rename the appropriate binary to **unphased**. Type **chmod a+x unphased** in the **bin/** directory to ensure that the file has executable permissions.

Starting UNPHASED

You can run UNPHASED either through a graphical user interface (GUI) or by typing into a command shell. The GUI allows you to select options from menus and hold the results of multiple analyses on screen at the same time. However the command line interface has greater flexibility, for example for writing shell scripts or piping the output.

To start the GUI, double click on the **unphased.jar** icon, or type **java -jar unphased.jar** in a command shell. A blank window will then appear with various menus for setting options. After loading the input files using the **File** menu and selecting options, start an analysis by selecting **Run:Start**. A new tab will appear holding the output of this run. The run can be interrupted by pressing control-C. You can edit the output at any time, save it to a file, and print it. You can have several analyses running at the same time, though this may make each analysis takes longer.

Selection of options from the menus is intuitive. When you are prompted to enter a haplotype, enter the alleles separated by spaces, not hyphens. If genotype analysis is

selected, enter the two alleles at each locus separated by a space, with the lower numbered allele first, and then a space to the next locus, and so on.

The GUI may not know where to find the main executable for UNPHASED, depending on your system configuration. In this case, select **Run:Locate executables** and then navigate to the directory containing the **unphased** or **unphased.exe** file. Double click on that file and the GUI will then remember where it is.

A few options are not directly available in the GUI. These can be invoked by selecting **Run:Command line arguments** and then typing in the command line syntax, as described below.

You can run UNPHASED from the command line in a unix shell or a Windows command prompt. Type **unphased** followed by a list of *options*. Each option consists of a dash followed by the option name. You do not need to type the option name in full, just enough letters to identify the option uniquely. Each option is followed by a list of *arguments*. Depending on the option, there may be no argument, one argument or several arguments. When several arguments are expected, each successive word on the command line is read as an argument until a word beginning with a dash is found, which is then parsed as the next option.

If a new option is expected but the next word does not begin with a dash, that word is interpreted as a pedigree file for input. You can list any number of pedigree files; they will be analysed one after the other in turn. To force UNPHASED to read a pedigree file, used the **-pedfile** option. This is useful if you use a variable argument option such as **-marker** and then want to add another pedigree file.

To send the output to a file instead of the screen, use the **-output** option followed by a file name. Alternatively, unix redirection will work as normal.

Here are a couple of examples of starting UNPHASED from the command line:

unphased mypeds.ped -marker 1 2 3 -missing -permutation 10

This will analyse the first three markers from the pedigree file **mypeds.ped**, estimate missing genotypes, and will run a permutation test with ten replicates.

unphased mypeds.ped -permutation 10 morepeds.ped

This will first analyse the file **mypeds.ped** and then the file **morepeds.ped**. Since the **-permutation** option has one argument, the word after that is interpreted as a pedigree file.

Some arguments take a haplotype as input. If more than one marker is being analysed, specify the haplotype as a list of alleles with a space between each one. If genotype analysis is selected, list the alleles as pairs with the lower value first. So for example

unphased myped.s.ped –window 2 –reference 1 2

will analyse two-locus haplotypes with the 1-2 haplotype as reference, and

unphased myped.s.ped –window 2 –genotype –reference 1 2 1 1

will analyse two-locus genotypes, using as reference the two-locus genotype with 1/2 at the first locus and 1/1 at the second.

Input files

The main input file for UNPHASED is a *pedigree file* containing all the subject trait and genotype data. There are several other input files that are optional. A *data file* specifies the format of the pedigree file. A *map file* gives the names of the markers in the pedigree file. A *binary pedigree file* is a more compact format for the genotype data. An *option file* lets you load a predefined set of options. Finally, a *marker list file* gives a list of markers to be analysed in turn.

Pedigree file

The pedigree file contains information on family relationships between study subjects, and all of the genotype and covariate data. UNPHASED uses the *LINKAGE pre-makeped* format, which in its simplest form consists of one line per subject, with columns organised as follows

PedID SubID PaID MaID Sex Trait M1A1 M1A2 M2A1 M2A2 ...

where:

PedID is a pedigree identifier which can consist of letters and numbers

SubID is a subject identifier which can consist of letters and numbers

PaID is the identifier of this subject's father, 0 if the father is not in the file

MaID is the identifier of this subject's mother, 0 if the mother is not in the file

Sex is 1 for male subjects, 2 for females, 0 for unknown

Trait contains the trait information for this subject. For a disease affection status, it is 1 for unaffected, 2 for affected, and 0 for unknown. For quantitative and other traits, the trait value is entered; if the value is unknown, enter a dash -. NB some other programs use "-9" as the missing value code: be careful to convert these to "-" before using UNPHASED.

M1A1 and **M1A2** are the two alleles for the first marker. Alleles can be any integer less than 16, or one of the letters A/C/G/T. Missing alleles are coded as 0 or N. If one marker is coded as letters, all other markers must be also. For sex-linked markers in

males, either enter the one allele followed by 0, or enter the allele as a homozygote genotype.

M2A1 and **M2A2** are the two alleles for the second marker, and so on.

For unrelated subjects, such as case/control data, each subject is regarded as a pedigree with one member. The **PedID** must be unique, the **SubID** can have any value, and both **PaID** and **MaID** must be 0.

If the pedigree file contains extended pedigrees, they will be broken up into nuclear families which are then treated as independent.

It is sensible to group members of the same pedigree together in the file, but it is not compulsory. When combining data sets into a single pedigree file, be careful that pedigree identifiers are not duplicated.

As far as possible, UNPHASED is "robust" to errors in the pedigree structure. This means that if a Mendelian inconsistency is encountered, a warning is printed and the pedigree is discarded, but the program continues to run. If a parent is missing, which may seem to happen if there is typo in **PaID** or **MaID**, a dummy parent will be created and no warning will be given. Sometimes this is an advantage, sometimes not. Therefore I strongly recommend that you run your file through PEDCHECK before running UNPHASED. The error checking in UNPHASED is reasonable but not exhaustive.

The first five columns of the pedigree file are mandatory. The above format is the default for the trait and genotype data, but other formats can be specified by using a data file, described below.

Data file

The data file, which is optional, specifies the structure of the pedigree file and the names of the traits and markers. If no data file is given, the format is assumed to be as above, and UNPHASED automatically determines whether the trait is binary or quantitative by looking at the values seen in column 6. If a different format is used, for example to include covariates, a data file is needed to define the organisation of columns. UNPHASED recognises two formats: *QTDT* format and *LINKAGE* format. It will automatically recognise which format has been used. In all cases the first five columns of the pedigree file are the same as above.

QTDT format is easier to work with. Each line of the file refers to a column in the pedigree file, in the same order. The first word defines the type of data and the second gives the name of the item. The type of data must be one of four codes:

M denotes marker genotype data. In this case the item refers to two columns of the pedigree file.

T denotes quantitative trait data.

C denotes covariate data. In UNPHASED this is interchangeable with quantitative trait data.

A denotes affection status data.

A full description of this format is available at

<http://www.sph.umich.edu/csg/abecasis/QTDT/docs/data.html>

Note that UNPHASED only recognises four codes and use of the **Z** and **S** codes will result in an error.

The following example is adapted from Gonalo Abecasis' web page referenced above. If the data file is the following

```
T HEIGHT
M GH1_SNP132
M GH1_SNP146
C AGE
```

then the pedigree file might look like this:

```
1000 1 0 0 1 1.87 1 2 2 2 40
1000 2 0 0 2 1.65 1 2 1 1 38
1000 3 1 2 1 1.80 1 1 1 2 20
1000 4 1 2 1 1.75 2 2 1 2 17
```

LINKAGE format is more complex, being designed originally for parametric linkage analysis. UNPHASED will insist on all elements of the file being present, but will only use the locus types and locus names, which are specified in the same way as for GENEHUNTER, following a # symbol after the number of alleles. A full description of the format is at

<http://linkage.rockefeller.edu/soft/linkage/sec2.6.html>

and also at

<http://linkage.rockefeller.edu/soft/gh/loadmark.html>.

To code either quantitative or covariate data, UNPHASED recognises locus code 4 in the same way as GENEHUNTER, that is a single line containing "4 0" is sufficient. More detail on this is at

<http://linkage.rockefeller.edu/soft/gh/vc.html>

A LINKAGE datafile for the pedigree file above could then look like this:

```

4 0 0 5 << first header line must have total number of loci
0 0 0 0 << second header line is ignored
1 2 3 4 << third header line can just count the loci
4 0 # HEIGHT
3 2 # GH1_SNP132
0.5 0.5 << allele frequencies, ignored by UNPHASED
3 2 # GH1_SNP146
0.5 0.5
4 0 # AGE
0 0 << interference code, ignored by UNPHASED
0.5 0.5 0.5 << recombination fractions, must be one fewer
than number of loci, but ignored by UNPHASED
1 0.1 0.5 << MLINK parameters, ignored by UNPHASED

```

When trait and marker names are defined by the data file, they must be used as the arguments to options. If no names are defined, traits and markers are automatically named 1, 2, ... from left to right in the pedigree file. Separate numerical lists are maintained for marker genotypes, affection statuses and quantitative traits.

Map file

If the pedigree file is in the default format, another way to specify the names of the markers is by using a map file. This file is in the *PLINK* format, in which each line corresponds to a marker, with columns organised as follows

Chr Name Dist Posn

where:

Chr is the chromosome number (this is ignored by UNPHASED)

Name is the marker name

Dist is the genetic map location (this is ignored by UNPHASED)

Posn is the physical map location (this is ignored by UNPHASED)

For biallelic markers, two further columns may be added, which give the names of the minor and major alleles respectively. This is necessary if the binary format is used. More detail on the *PLINK* format is at

<http://pngu.mgh.harvard.edu/~purcell/plink/data.shtml#map>

Phenotype file

Another way to specify additional phenotypes for subjects is in a phenotype file. This is in the *PLINK* format, with columns organised as follows

PedID SubID Pheno1 Pheno2 ...

where:

PedID is a pedigree identifier which can consist of letters and numbers

SubID is a subject identifier which can consist of letters and numbers

Pheno1 is the value of the first phenotype

Pheno2 is the value of the second phenotype

and so on. A header line can be optionally included, which has the form

FID IID Name1 Name2 ...

where:

FID must be stated literally and in uppercase

IID must be stated literally and in uppercase

Name1 is the name of the first phenotype

Name2 is the name of the second phenotype

and so on. The given names of the phenotypes are then used as arguments to options. If no header line is given, the phenotypes are named 1, 2, ... and so on. More detail on the *PLINK* format is at

<http://pngu.mgh.harvard.edu/~purcell/plink/data.shtml#pheno>

Note that all phenotypes are treated as quantitative traits. Affection status data can only be encoded in the pedigree file. Missing phenotypes are coded by a dash – . NB some other programs use “-9” as the missing value code: be careful to convert these to “-” before using UNPHASED.

If a data file is specified as well as a phenotype file, the phenotypes in the phenotype file are appended to those given in the data file.

Binary pedigree file

UNPHASED supports the *PLINK* binary format. This format requires three files. The first, called the *.fam* file consists of the first five columns of an ordinary pedigree file. In UNPHASED the *.fam* file is used wherever an ordinary pedigree file is expected. The second, called the *.bim* file is a map file as described above, with the two extra columns for the allele names. The third file has the suffix *.bed* and contains the genotypes encoded in a compressed format. This file is referred to as the *binary pedigree file* throughout this document. Full details of the binary format are at

<http://pngu.mgh.harvard.edu/~purcell/plink/data.shtml#bed>

Option file

Instead of selecting the program options from the menus, or typing them on the command line, UNPHASED can read its options from a previously created *option file*. This file contains one line per option, with the arguments for each option listed on the same line. The leading dash can be omitted from the option name. For example, the option file:

```
-window 2  
-genotype  
reference 1 2 1 1
```

will set up the same options as the command line

```
unphased -window 2 -genotype -reference 1 2 1 1
```

Only one option file should be specified per run. When using an option file, additional options can be specified on the command line or through the GUI.

Marker list file

This file is used to specify a series of analyses of different markers. While a single command can be used to analyse several markers (see the **-marker** and **-window** options below), the marker list file allows you to analyse haplotypes of different lengths, with different conditioning and tag markers, within a single run, and then do permutation tests over the whole set of analyses.

Each line of the marker list file contains command line options specifying the test, conditioning and tag markers for one analysis. For example, the marker list file

```
-marker 1 -condition 2 -tag 3  
-marker 2 -tag 1  
-marker 1 2  
-marker 1 2 -window 2
```

will first analyse marker 1, with marker 2 as the conditioning marker and marker 3 as the tag marker. Then it will analyse marker 2 with marker 1 as the tag marker. Then it will analyse marker 1, followed by marker 2. Finally it will analyse markers 1 and 2 as a two-marker haplotype.

Output

Screen output

The screen output is in three sections. The first section lists all the analysis options and selected covariates, if any. This section can be shortened using the **–brief** option. The second section displays the progress of the analysis. The number of likelihood evaluations at each stage is shown. The third section gives various tables of results for each haplotype (or allele or genotype), for the currently selected markers.

Firstly, the marginal haplotype frequencies are shown. One table is shown for the families in the pedigree file, and another for the unrelated subjects. The tables contain the following information:

Allele/Haplotype/Genotype: the genetic unit of interest. Alleles at successive loci are separated by a dash –. For genotypes, alleles at the same loci are separated by a slash /. Thus, 1/2–3/4 means genotype 1/2 at the first locus and 3/4 at the second.

Trans/Case: the estimated count of this haplotype in affected offspring or cases. Note that this count includes contributions from homozygous parents: TDT-like counts of transmitted and untransmitted alleles are not available, as they are hard to define when parents are missing.

Untrans/Control: the estimated number of non-transmissions of this haplotype, or the estimated number in controls. In nuclear families, this also includes transmissions to unaffected children. Note that this count includes contributions from homozygous parents: TDT-like counts of transmitted and untransmitted alleles are not available, as they are hard to define when parents are missing. In genotype analyses, the non-transmitted genotypes are the three genotypes that could be formed from the parental haplotypes, but were not transmitted.

Count: For quantitative traits, the Trans/Case and Untrans/Control columns are replaced by a single column giving the total count of the haplotype.

T-Freq/Ca-Freq: the marginal frequency of the haplotype in affected offspring or cases. This is the count in the **Trans/Case** column divided by the column total.

U-Freq/Co-Freq: the marginal frequency of the haplotype among untransmitted haplotypes or controls. This is the count in the **Untrans/Control** column divided by the column total.

MarFreq: For quantitative traits, the above two columns are replaced by a single column giving the marginal frequency of the haplotype over all subjects.

Note that these are marginal frequencies in the sense that they are averaged over all levels of all covariates. UNPHASED cannot for example display the frequencies in males and females separately. This is because when there are several covariates the number of possible combinations can become very large and UNPHASED does not fully model all combinations.

Next the overall test of association is displayed. This is a global, or omnibus, test of whether any haplotypes are associated. The overall test of association is a likelihood ratio test. The log-likelihoods for the null and alternative hypotheses are displayed, the hypotheses themselves being determined by the analysis options. The **likelihood ratio** statistic is minus twice the difference in log-likelihoods, and is asymptotically distributed as χ^2 with degrees of freedom (**df**) equal to the difference in number of free parameters between the two hypotheses. The **p-value** is the probability of observing a likelihood ratio statistic at least as large as this one, if the null hypothesis were true.

Then the estimated haplotype effects are displayed in a table containing the following:

Odds-R: the estimated odds ratio for this haplotype. For quantitative traits this column is headed **AddVal** and shows the estimated additive genetic value for this haplotype, assuming a normally distributed trait and small deviations from the mean. These effects are shown relative to one haplotype, termed the *reference* haplotype. By default this is the first haplotype in sorted numerical order, but it can be changed with the **-reference** option. Odds ratio has the usual meaning for case-control data; for family data it is actually the relative risk, which equals the odds ratio for a rare disease. Additive value gives the change in expected trait value due to this haplotype, relative to the reference haplotype. So if a subject has haplotypes with additive values x and y , its expected trait mean is $x+y$ higher than for a subject homozygous for the reference haplotype. The reference haplotype has additive value zero, compared to itself. The default model assumes a normal trait distribution with unit variance, with small additive values. For large additive values, the **-normal** option permits more accurate estimation, but see the documentation for this option before using it. Non-unit variance is specified by the **-variance** option. More details of quantitative trait analysis are in the “Methods” section.

When effect modifiers are included, the odds ratio and additive values apply at the baseline levels of the modifiers, or at a value of zero for continuous modifiers.

95%lo: 95% lower confidence limit for the odds ratio or additive value.

95%hi: 95% upper confidence limit for the odds ratio or additive value.

If individual haplotype tests are selected with the **-individual** option, two more columns will appear:

Chisq : a score statistic for the effect of this haplotype relative to all others pooled together. Note that the odds ratios, additive values and their confidence intervals are relative to the reference haplotype, whereas this score statistic is relative to all other

haplotypes pooled together. It can happen that this statistic is significant when the confidence interval for the odds ratio overlaps 1, and vice versa. To get odds ratios relative to other haplotypes pooled together, use the **–specific** option.

P-value : the asymptotic significance of the score statistic referring to χ^2 on 1df.

If covariates are included, this table refers to the effects in subjects having the baseline level for all factorial covariates and a value of 0 for all continuous covariates. Further tables are now displayed, one for each level of each covariate. For continuous covariates, one table is displayed corresponding to a covariate value of 1. The important point about the covariate outputs is that they reflect differences from the baseline covariate level, rather than absolute values. For example, if sex is used as a covariate, with females as the baseline, then the main table refers to effects in females whereas the next table refers to differences between males and females. For a continuous covariate, such as age, the main table refers to effects at age 0 whereas the next refers to differences at age 1 from age 0.

For confounders, the **Offset** column gives a parameter which models a change in haplotype frequency compared to that at the baseline covariate level. For effect modifiers, the **Odds-R** or **AddVal** columns give the odds ratio or additive value compared to that of the reference haplotype, after allowing for the baseline effects in the main part of the output. Confidence intervals are as above, and χ^2 tests for the covariate effects can be computed. These test whether the haplotype effects are equal for each covariate level, in other words whether there is any gene-covariate interaction after allowing for effects at the baseline level.

If LD output is selected, further tables are displayed consisting of D' and r^2 values for each haplotype, together with a global D' statistic. Separate tables are shown for families and unrelateds, and for transmitted/case and untransmitted/control haplotypes.

Haplotype dump

UNPHASED allows all the possible haplotypes consistent with each family to be written to a file together with the probability of each haplotype configuration. This *dump file* can be useful although it is not particularly easy to read.

For nuclear families, each line of the dump file corresponds to a family. If all offspring are considered jointly (the default option), the subject IDs of these siblings are first listed. Then each configuration is given as a list of phased haplotype pairs with loci separated by a dash –, haplotypes separated by a slash /, transmitted and untransmitted haplotypes separated by a backslash \, and siblings separated by a comma , . Then the probability of the configuration is given. Note that the dump file distinguishes the phase of each configuration, so that some unphased configurations seem to appear more than once. For example, the single locus heterozygote will appear both as 1/2 and 2/1.

If offspring are considered independently (the **–nolinkage** option), each sibling is listed in turn with the same syntax as above.

For unrelated subjects, the configurations are listed as phased haplotype pairs with their corresponding probabilities.

Tabular output

Using the **–tabularfile** option, UNPHASED can create output files in a tabular format that can be read easily by programs such as R, EXCEL and HAPLOVIEW, for easy browsing and graphical display of results. The format is similar to the haplotype association output of PLINK, with minor modifications. Separate files are created for families and for unrelated subjects, but the odds ratios (or additive values) and p-values are from the combined analysis. Separate files are also created for each trait analysed within a single run. The output files can be viewed in the HAPLOVIEW program by selecting “PLINK format” for the input, with the “Non-SNP” option.

Options

UNPHASED provides options for performing several different types of analysis. Here the options are grouped according to the menus in the GUI, with the corresponding command line arguments. At the end there is a full list of command line options together with their minimal abbreviated forms.

Reading and writing files

The **File** menu is used for input of data files and output of results, and also for managing the appearance of the GUI.

Open pedigree file

–pedfile <filename1> <filename2> ...

Reads in the pedigree file from disk. The format is described above in the section “Input files”.

Open data file

–datafile <filename>

Reads in the data file from disk. The format is described above in the section “Input files”.

Open map file

–mapfile <filename>

Reads in the map file from disk. The format is described above in the section “Input files”.

Open phenotype file

–phenofile <filename>

Reads in the phenotype file from disk. The format is described above in the section “Input files”.

Open binary pedigree file

–bedfile <filename>

Reads in the binary pedigree file from disk. The format is described above in the section “Input files”. If this format is used, the *.fam* file should be specified as the pedigree file, the *.bim* file as the map file, and the *.bed* file as the binary pedigree file.

Open option file

–optionfile <filename>

Reads in the option file from disk. The format is described above in the section “Input files”. If selected in the GUI, options that specify input and output files are ignored, and these must be set up explicitly within the GUI.

Close pedigree file

Closes the currently loaded pedigree file in the GUI. No more analyses can be performed until another pedigree file has been opened.

Close data file

Closes the currently loaded data file in the GUI. This means that all currently loaded marker and trait names will be lost and the GUI will assume that the currently loaded pedigree file has the basic format.

Save output

–output <filename>

In the GUI this saves the currently displayed tab to a text file. On the command line, it sends the main output to a file instead of the screen.

Save options

This is only available in the GUI, and creates a new option file containing the current selection of options. Options that specify input or output files are omitted.

Rename tab

In the GUI, tabs are automatically named after the pedigree file, and also display the count of how many analyses have been performed so far. This action lets you change the name of the currently displayed tab.

Close tab

In the GUI, this closes the currently display tab.

Print

In the GUI, this attempts to print the currently selected tab. If this doesn't work, it is more likely to be a problem with your local Java printing setup than with the GUI itself. Please check this out before reporting a bug.

Selecting traits to analyse

The **Trait** menu selects which traits to analyse. If the pedigree file has the basic format, there is no need to use this menu. UNPHASED distinguishes between two types of trait: binary traits taking values 1 or 2 (also called affection status or disease), and quantitative traits that can take any numerical value. Several traits can be analysed in one run, with each trait being analysed in turn. If both binary and quantitative traits are selected, UNPHASED analyses all the binary traits first, and then all the quantitative traits. If no traits are selected, UNPHASED analyses just the first binary trait if one is present, otherwise the first quantitative trait.

Affection status

-disease <name1> <name2> ...

Specifies which binary traits to analyse. Each trait will be analysed in turn. In the GUI they will be analysed in the lexical order, whereas on the command line they will be analysed in the order in which they are listed.

Quantitative trait

-trait <name1> <name2> ...

Specifies which quantitative traits to analyse. Each trait will be analysed in turn. In the GUI they will be analysed in the lexical order, whereas on the command line they will be analysed in the order in which they are listed.

Selecting markers to analyse

The **Marker** menu selects markers for analysis. UNPHASED distinguishes between three types of marker. *Test markers* are the markers of interest, which will be tested for association. *Conditioning markers* may themselves be associated, and those associations are factored out of the association of the test markers. *Tag markers* can add information about missing genotypes, but any associations are not considered. For each haplotype, UNPHASED estimates a population frequency, which depends on the alleles at all three sets of markers, and an odds ratio, which depends only on the alleles at test and conditioning markers.

When displaying haplotypes in the output, UNPHASED places the conditioning markers first, then the test markers, then the tag markers. So if there is one of each type of marker, the haplotype 1-2-3 refers to allele 1 at the conditioning marker, allele 2 at the test markers and allele 3 at the tag marker.

When several analyses are specified by the **-window** option below, the conditioning and tag markers are held fixed while the test markers are varied.

Test markers

-marker <name1> <name2> ...

Selects a subset of markers out of the pedigree file. Each marker will be tested in turn for association. Use the names in the data file, or if no data file is loaded, use numbers 1, 2, ...

Each marker will be analysed in turn. In the GUI they will be analysed in the lexical order, whereas on the command line they will be analysed in the order in which they are listed. To analyse multi-locus haplotypes, set a window size as described below.

If no markers are selected, UNPHASED automatically selects all the markers in the file. This gives a quick way to specify that every marker should be analysed.

Conditioning markers

–condition <name1> <name2> ...

Suppose the test markers are in linkage disequilibrium (LD) with other markers that are known to be associated. We want to know if the test markers are directly associated, beyond the indirect association occurring through LD. In order to perform such a test, the previously associated *conditioning markers* should be selected using this option.

The conditioning markers are held fixed over a sliding window analysis. In the output, the alleles of the conditioning markers are displayed to the *left* of the test markers. This convention also applies to options that have a haplotype as an argument (for example –reference).

Tag markers

–tag <name1> <name2> ...

When some genotypes are missing, information from other markers can be used to estimate them, if the markers are in linkage disequilibrium. Here these additional markers are called *tag markers*.

The tag markers are held fixed over a sliding window analysis. In the output, the alleles of the tag markers are displayed to the *right* of the test markers. This convention also applies to options that have a haplotype as an argument (for example –reference).

Window size

–window <value>

Used to specify multi-locus analysis. The window size is the number of test markers included in each multi-locus analysis.

By default, UNPHASED runs a sliding window analysis over the selected markers. For example, if the selected test markers are 1, 2, 3, 5, 8, and the window size is 3, then the following 3-marker analyses will be performed:

Markers 1, 2, 3

Markers 2, 3, 5

Markers 3, 5, 8

Instead of running a sliding window, UNPHASED can analyse all combinations of markers with the given window size; see below.

Conditioning and tag markers are held fixed over multiple test markers selected by the sliding window.

If no window size is specified, the default value is 1.

All marker combinations

–allcombinations

Specifies that all combinations of markers with the given window size should be analysed, instead of running a sliding window over the test markers.

For example, if the selected test markers are 1, 2, 3, 5, 8, and the window size is 3 with this option selected, then the following 3-marker analyses will be performed:

Markers 1, 2, 3

Markers 1, 2, 5

Markers 1, 2, 8

Markers 1, 3, 5

Markers 1, 3, 8

Markers 1, 5, 8

Markers 2, 3, 5

Markers 2, 3, 8

Markers 3, 5, 8

All window sizes

–allwindows

Specifies that each window size will be run in turn, starting with 1 and ending with the number of test markers. This option overrides the **–window** option. It can be used in conjunction with the **–allcombinations** option.

Marker list file

-listmarkerfile

Reads a series of markers to analyse from the given file. The syntax is specified above in the section “Input files”. If a marker list file is used, it overrides any analyses specified with the **–marker**, **–condition** and **–tag** options.

Selecting covariates

The **Covariate** menu selects covariates to be included in the analysis. UNPHASED allows modeling of interactions between haplotypes and environmental or other covariates. There is a distinction between two types of covariate. *Confounders* are covariates whose values influence the population frequency of a haplotype and possibly the disease prevalence or trait mean. An example might be geographical region. *Modifiers* are covariates whose values influence the odds ratio of a haplotype, or its additive value on a quantitative trait: that is, they are effect modifiers. A common example is sex. UNPHASED allows for the possibility that an effect modifier also influence the population haplotype frequencies.

UNPHASED conditions on all trait and covariate values (see the “Methods” section). This means that main effects of covariates are conditioned out of the likelihood and they cannot be estimated. In other words, all trait values are automatically adjusted for the main effect of a covariate, and we do not, for example, need to worry about adjusting measurements for age unless there is a confounding effect or a gene-covariate interaction.

A confounder or modifier can be either continuous (the default) or a *factor*. Factors are discrete variables taking values among a finite set of *levels*. Each factor has a *baseline* level, which is the level for which UNPHASED gives the main analysis. Once a covariate is defined as a factor, UNPHASED automatically reads its levels from the pedigree file. The levels can be any numerical values.

Confounders

–confounder <name1> <name2> ...

Any number of confounders can be selected. They are assumed to have effects on haplotype frequency but not haplotype risk. See the “Methods” section for exactly how they are modeled by UNPHASED.

Parental and offspring sex can be used as confounders, although it is unlikely that haplotypes will have different population frequencies between males and females. To use parental sex as a confounder, use the name “parsex”. To use offspring sex as a confounder, use the name “sibsex”. For unrelated subjects, also use “sibsex”.

Modifiers

–modifier <name1> <name2> ...

Any number of modifiers can be selected. They are assumed to modify the effect of haplotypes, that is the odds ratio or mean. See the “Methods” section for exactly how they are modeled by UNPHASED.

Parental and offspring sex can be used as modifiers. To use parental sex as a modifier, use the name “parsex”. This corresponds to a parent of origin effect in the offspring. To use offspring sex as a modifier, use the name “sibsex”. For unrelated subjects, also use “sibsex”. This corresponds to differential risk between males and females.

Factors

–factor <name1> <name2> ...

Defines which covariates should be regarded as factors with discrete levels. If parental or offspring sex are used as covariates, they are automatically set as factors with two levels and they should not be included in this option.

Baselines

–baseline <value1> <value2> ...

The argument for this option should be a list of baseline levels with one for each selected covariate. List the confounders first, then the modifiers. For covariates that are not factors, the baseline level will be ignored and can be given as 0. For parental and offspring sex, a baseline level of 1 corresponds to males and 2 to females.

Setting up the type of analysis

The **Analysis** menu sets up the main analysis. The first part of the menu selects an analysis “model”, which is one of six built-in designs for coding haplotype risks. The next part allows additional tests to be reported apart from the main likelihood ratio test. The other parts define particular haplotypes for various purposes.

The analysis models are described in further detail in the Methods section of this manual.

Full model

–model full

UNPHASED has several different parameterisations for the haplotype risk, which can be used to perform various different tests. The full model is the default model and is used when the **–model** option is omitted. It corresponds to full haplotype coding, with a unique odds ratio parameter for each haplotype. Under the null, all the odds ratios are equal. With conditioning markers, the null defines odds ratios to be equal for all haplotypes sharing the same conditioning alleles, but haplotypes with different conditioning alleles can have different odds ratios. Among the conditioning tests, this has the most degrees of freedom, so will tend to be less powerful unless there are strong interactions between test and conditioning markers.

When specific test and conditioning haplotypes are given, the inference only applies to full haplotypes that match those sub-sections. Odds ratios for other haplotypes are considered as nuisance parameters. The constraints on odds ratios are as follows:

H_0 : all haplotypes matching the specific conditioning haplotype have equal odds ratio. All other haplotypes have unique odds ratios.

H_1 : all haplotypes matching the specific conditioning haplotype and not matching the specific test haplotype have equal odds ratio. All other haplotypes have unique odds ratios.

In the above, “matching” is defined as true if no specific haplotype is given. “Unique” applies only to the conditioning and test haplotypes: if there are tag markers, then haplotypes with the same conditioning and test haplotypes but different tag haplotypes will have the same odds ratio.

Haplotype main effects

–model **haplomain**

In this model, the odds ratio of a haplotype is modeled by multiplicative contributions from the conditioning haplotype and from the test haplotype. The null hypothesis is that none of the test haplotypes make any contribution, in other words they have no main effect on the overall odds ratio. This test has fewer degrees of freedom than the full model, but may be less likely to detect an effect when there is strong interaction between conditioning and test haplotypes.

If no conditioning markers are selected, this model will give the same results as the full model.

As for the full model, specific test and conditioning haplotypes restrict the inference to full haplotypes that match those sub-sections. Odds ratios for other haplotypes are considered as nuisance parameters. The constraints on odds ratios are as follows:

H_0 : all haplotypes matching the specific conditioning haplotype have equal odds ratio. All other haplotypes have unique odds ratios.

H_1 : all haplotypes matching the specific conditioning haplotype and not matching the specific test haplotype have equal odds ratio. Haplotypes matching the both the specific conditioning haplotype and the specific test haplotype have odds ratios estimated by the haplotype main effects model. All other haplotypes have unique odds ratios.

Allele main effects

–model **allelemain**

In this model, the odds ratio of a haplotype is modeled by multiplicative contributions from each test marker. The null hypothesis is that none of the test markers make any

contribution to the overall odds ratio. This test has fewer degrees of freedom than the previous models, but may be less likely to detect an effect when there is strong interaction between markers.

Conditioning markers are modeled by full haplotype coding, so the main effects only apply to test markers. Because the test markers are modeled by main effects, there is no sensible interpretation of a specific test haplotype, and the **–specific** option is ignored by this model. However a specific conditioning haplotype may still be used to restrict inference to a subset of haplotypes. The constraints on odds ratios are as follows:

H_0 : all haplotypes matching the specific conditioning haplotype have equal odds ratio. All other haplotypes have unique odds ratios.

H_1 : all haplotypes matching the specific conditioning haplotype and not matching the specific test haplotype have equal odds ratio. Haplotypes matching both the specific conditioning haplotype and the specific test haplotype have odds ratios estimated by the allele main effects model. All other haplotypes have unique odds ratios.

Common main effect

–model commonmain

This model is the same as the allele main effects model, except that the effects of all test markers are constrained to be equal. This is equivalent to treating the total number of minor alleles carried as a single risk factor (assuming that minor alleles are coded identically across markers). As there is only one degree of freedom, this test can be the most powerful among all the models when all the markers have a similar effect size. This is a currently popular test for the combined effect of several rare variants in a region.

Gene-gene interaction

–model gxg

This model tests for gene-gene interaction by considering whether the odds ratios of the conditioning haplotypes are independent of those of the test haplotypes. That is, it compares the odds ratio of the full haplotype to that expected if the conditioning and test haplotypes had independent risks. Note that this only tests for *cis*-phase interaction, unless genotype tests are selected, in which case either *cis*- or *trans*-phase interaction could be detected. Note also that this statistical definition of interaction does not necessarily correspond to the classical model of epistasis.

If no conditioning markers are selected, this model will compute identical likelihoods under both null and alternative.

Essentially, this model compares the likelihoods between the haplotype main effects model described above, and the full model. Inference can be restricted to haplotypes

matching specific conditioning and test haplotypes. More precisely, the constraints on odds ratios are as follows:

H_0 : all haplotypes have odds ratios estimated by the haplotype main effects model.

H_1 : haplotypes matching both the specific conditioning haplotype and the specific test haplotype have unique odds ratios. All other haplotypes have odds ratios estimated by the haplotype main effects model.

Null model

–model null

This option only evaluates the null model in which all haplotypes have the same effect. The option is useful for getting rough estimates of haplotype frequencies, and also for quickly getting results for the individual haplotype tests (see **–individual** below) without computing a global test.

Individual haplotype tests

–individual

This option will give individual tests for each haplotype in turn. The test is a score test for a difference in risk between a haplotype and all the others pooled together. This is in contrast to the odds ratios, which are shown relative to a single reference haplotype. It is possible for the individual test to be significant when the confidence interval for the odds ratio includes 1, and vice versa. To get an estimate of a haplotype risk relative to all others together, see the “Specific test haplotype” section.

This option has changed from the previous version of UNPHASED, 2.4, in that it now uses a score test whereas the old version used a likelihood ratio test. Results should be very similar but there may be differences in some cases. The likelihood ratio can still be used to double check results from the score test, again see the “Specific test haplotype” section.

The individual haplotype tests take account of the **–tag** option but not the **–condition** option. That is, when tag markers are selected, the individual test only applies to the test markers, with the tag markers still used to estimate missing data. However, when there are conditioning markers, the individual test applies to the haplotype consisting of both conditioning and test markers.

Test confounder effects

–testconfounders

For each level of a confounder, this gives a global Wald test of whether any of the regression coefficients are non-zero.

Test modifier effects

–testmodifiers

For each level of a modifier, this gives a Wald test for each haplotype, indicating whether the modifier changes its odds ratio or mean when compared to the baseline level.

Compare haplotype risks

–compare <allele1> <allele2> ...

–with <allele1> <allele2> ...

This option allows the risks of two haplotypes to be compared. Under the null, their risks are set to be equal while all other haplotypes have freely estimated risk. Under the alternative all haplotypes can have different risks. The two haplotypes are specified with the **–compare** and **–with** options respectively.

Wildcards are allowed in the pairwise comparison, and are specified by 0. These have the effect of forcing the odds ratios to be equal for all haplotypes matching the wildcard. So for example

unphased myped.s.ped –window 2 –compare 1 0 –with 2 2

will test whether all haplotypes with a 1 at the first locus have the same odds ratio as the 2-2 haplotype. If there are wildcards in both haplotypes, a haplotype is considered to match the second haplotype only if it does not match the first.

The comparison options override the analysis specified by the **–model** option.

Reference haplotype

–reference <allele1> <allele2> ...

This option specifies the reference haplotype, which is defined to have odds ratio 1. The default reference haplotype is the first one when the haplotypes are sorted into numerical order.

Specific test haplotype

–specific <allele1> <allele2> ...

This option restricts the inference to a specific test haplotype. How this is done depends on which model is selected, and is described in the sections above. The specific haplotype should only be specified for the test markers. If conditioning or tag markers are selected, the number of arguments to this option must remain as the number of test markers.

Wildcards are allowed as arguments and are specified as 0.

Specific conditioning haplotype

–condspecific <allele1> <allele2> ...

This option restricts the inference to a specific conditioning haplotype. How this is done depends on which model is selected, and is described in the sections above. The specific haplotype should only be specified for the conditioning markers. If test or tag markers are selected, the number of arguments to this option must remain as the number of conditioning markers.

Wildcards are allowed as arguments and are specified as 0.

Analysis options

The **Options** menu sets up various options for performing the analysis.

Missing data : Certain haplotypes only

–certain

Restricts analysis to subjects for whom the haplotypes are unambiguous. In nuclear families this corresponds to at most one heterozygous intercross. In unrelated subjects this selects only homozygous subjects, unless only one marker is selected. In all cases, subjects with missing genotypes are discarded.

Missing data : Uncertain haplotypes

This is the default setting and includes all subjects that have no missing genotype data.

Missing data : Uncertain haplotypes and missing genotypes

–missing

Selects all subjects that have some genotype data, and averages over all possible completions of the data. It is the slowest of the three missing data options but uses the most information.

Rare haplotypes : Rare frequency threshold

–rare <value>

Sets a frequency below which haplotypes are designated as rare. First, UNPHASED estimates all the haplotype odds ratios and estimates their marginal frequencies. Then it identifies the rare haplotypes, and from then on constrains them to have the same odds ratio, in order to reduce computation. The pooled odds ratio is treated as a nuisance parameter in the likelihood ratio tests, so that rare haplotypes are not tested for association.

By default the haplotype must be rare in both cases and controls. This can be changed with the **–rarfreq** option.

If the value is greater than 1, it is interpreted as a percentage, unless the **–cellcount** option is selected.

Rare haplotypes : Zero frequency threshold

–zero <value>

Sets a frequency below which haplotypes are assumed to be absent from the population. First, UNPHASED estimates all the haplotype frequencies and odds ratios. Frequencies below this threshold are rounded down to zero. From then on, configurations involving these haplotypes are ignored. This can significantly speed up the computation and declutter the output.

By default the haplotype must be absent from both cases and controls. This can be changed with the **–userare** option.

The default value of the zero frequency threshold is 10^{-8} .

Rare haplotypes : Threshold on cell counts

–cellcount

Identifies rare haplotypes according to the cell counts rather than the marginal frequencies. This option does not affect the zero frequency threshold.

Rare haplotypes : Use frequencies in : Both cases and controls

–userare both

Defines a haplotype to be rare if its marginal frequency is below the threshold in both cases and controls. This is the default setting.

Rare haplotypes : Use frequencies in : Either cases or controls

–userare either

Defines a haplotype to be rare if its marginal frequency is below the threshold in either cases or controls.

Rare haplotypes : Use frequencies in : Just cases

–userare case

Defines a haplotype to be rare if its marginal frequency is below the threshold in the cases.

Rare haplotypes : Use frequencies in : Just controls

–userare control

Defines a haplotype to be rare if its marginal frequency is below the threshold in the controls.

Nuclear families : Assume no linkage

–nolinkage

When there are multiple siblings in a nuclear family, transmissions are not independent if there is known to be linkage in the region. When selected, this option assumes that there is no linkage, so that the siblings are assumed to be independent. This results in a faster and more powerful analysis than the default.

Nuclear families : Model odds ratio in parents

–parentrisk

This option enables more valid inference when there are missing parents and multiple siblings in a nuclear family (see the “Methods” section). However it introduces more nuisance parameters and therefore reduces power and increases running time.

Nuclear families : Use only one family-based control

–onefbc

In family based analyses, the transmitted genotype is usually compared to the three other genotypes that could have been transmitted by the parents. When this option is selected, only one genotype is used for comparison, being the one made up of the two haplotypes not transmitted by the parents. This may be appropriate for situations in which the parental transmissions are not independent, for example conditional tests with dominance at the conditioning loci.

Nuclear families : Unmatched family-based controls

–hhrr

This option uses the untransmitted haplotypes as family-based controls, as usual, but performs an unmatched analysis. This is similar to the original haplotype-based haplotype relative risk method (hence the name) and gives the same results as the TDTPHASE program in UNPHASED 2.4. It is less robust to population stratification than this version, but can be slightly more powerful. Furthermore, if extra unrelated controls are available, they can increase the power further since the unmatched analysis allows them to be readily included.

Nuclear families : Faster model for sibships

–sibship

This applies a special model to sibships with both parents missing and more than one trait value observed. It allows sibships to be analysed without the **–missing** option, and is faster than the standard model with **–missing**. There is a slight loss of power compared to the standard model, owing to an extra conditioning step.

Genetic : Genotype tests

–genotype

Selects genotype tests instead of haplotype tests. Genotypes are unphased throughout.

Genetic : Condition on genotypes

–condgenotype

Models genotypes at the conditioning markers but not at the test and tag markers. This can be useful if there is a strongly dominant effect at the conditioning markers.

The output will show estimates for combinations of genotypes and haplotypes, for example 1/2–1, 1/2–2, etc. This is confusing from a genetics point of view, but statistically it corresponds to using the conditioning genotype as a covariate when making inferences on test haplotypes.

Genetic : Autosome

This default option assumes that the markers are autosomal.

Genetic : Chromosome X

–chrX

Indicates that all markers are on chromosome X. In family studies, only the maternal transmissions are counted. In male subjects, only one chromosome is considered. If males and females are combined in the same analysis, the odds ratio in males is assumed to equal the heterozygote odds ratio in females. Unless this is expected to be the case, it is advisable to include “sibsex” as a modifier, in order to give separate odds ratios in males and females.

Genetic : Chromosome Y

–chrY

Indicates that all markers are on chromosome Y. In nuclear families, the parents will be discarded and just the first male offspring will be used. It will be then be analysed as an unrelated subject.

Quantitative trait : Model normal distribution

–normal

Uses an explicit normal distribution to model quantitative traits, instead of the small-effects approximation used by default (see the “Methods” section). This option is more accurate at estimating the additive value of a haplotype, particularly when the effect is large, and can be more powerful when the trait is truly normal. However, the likelihood can be more difficult to maximize numerically, and this model is less robust to non-normality than the logistic model.

Quantitative trait : Uncentred traits

–uncentred

If the traits are not zero-centred, some analyses require the trait mean to be estimated. This increases the running time and can lead to unstable estimates, so it is recommended that quantitative traits are standardized or at least zero-centred before analysis. If this is not the case, this option will ensure that the mean is estimated from the data. This option is only necessary if there are nuclear families and the **–parentrisk** option has also been selected: see the Methods for more details.

Quantitative trait : Trait variance

–variance

Specify the trait variance. When used without the **–normal** option, this does not change the p-value but does scale the estimated additive value. When used with the **–normal** option, this changes both the p-value and the additive value. Correct inference occurs only when the correct variance is used. However the variance may be difficult to estimate numerically, and may even not be identifiable in the likelihood, so for these reasons the variance is left as an input option. It could be profiled across several values to choose the most likely value.

Quantitative trait : Residual sib covariance

–covariance

Specify the covariance between siblings after the genetic variance has been removed. When used without the **–normal** option, this does not change the p-value but does change the estimated additive value. When used with the **–normal** option, this changes both the p-value and the additive value. Correct inference occurs only when the correct covariance is used. However the covariance may be difficult to estimate numerically, and may even not be identifiable in the likelihood, so for these reasons the covariance is left as an input option. It could be profiled across several values to choose the most likely value.

Permutation test

–permutation <value>

Specifies how many random permutations to run. In nuclear families, the permutations are generated by randomizing the transmission status of the parental haplotypes. In unrelated subjects, the trait values are randomly shuffled between subjects. The randomization is held constant over all analyses specified by the **–disease**, **–trait**, **–marker** and **–window** options, as well as any tests selected by the **–individual**, **–testconfounders** and **–testmodifiers** options. In each permutation, the minimum p-value is compared to the minimum p-value over all the analyses in the original data. This allows for multiple testing corrections over all tests performed in a run.

Show quantile from permutations

–quantile <value>

Outputs a percentile point from the permutation distribution of the minimum p-value. This allows us to say whether other results are also significant after correcting for multiple testing, apart from that with the minimum p-value. For example, if the second-most significant result from the original analysis is also less than the 5th percentile of the permutation distribution, we could say that it is also significant over multiple tests.

Convergence threshold

–epsilon <value>

Sets the tolerance for convergence in the numerical maximisation of the likelihood. The default value is 10^{-8} . This sometimes leads to convergence problems and a value of 10^{-6} can often be used without affecting results.

Random restarts

–restarts <value>

Maximises the likelihood several times, using randomly chosen starting points. It is useful if the likelihood seems not be maximised at the first attempt, for example if there is an extraordinary difference between null and alternative likelihoods, or the alternative is less than the null.

Random number seed

–randomseed <value>

Sets the initial seed for the random number generator. The default value is 1.

Output options

The **Output** menu sets various options for controlling the output of UNPHASED.

Brief output

–brief

When not selected, UNPHASED prints a list of all options and their currently selected values. This option stops this list from printing, making the output briefer.

Show LD measures

-LD

Displays table of D' and r^2 measures for each haplotype, together with global D' statistics. If more than two markers are selected, the LD is computed between the first marker and all the others grouped together. This can be used to assess how useful multiple markers could be when used with the **-tag** option.

Output permutation analyses

-permoutput

Displays the output from each permuted data set. It can be useful for observing the behaviour of the permutation distribution.

Output running time

-time

Displays the time in seconds taken to perform all the analysis.

Dump haplotypes to file

-dumpfile <filename>

Writes the probabilities of all possible haplotypes for each subject into a file. The format of the file is described in the section “Haplotype dump” above.

Just the most likely haplotypes

-mostlikely

Writes only the most likely haplotype solutions to the dump file, together with their probabilities. There may be more than one most likely solution.

Output in tabular format

-tabularfile <filename>

Creates files of tabular output that can be conveniently viewed in programs such as R, EXCEL, and HAPLOVIEW. Two files are created for each trait analysed, one for the family data with name `<filename>.<trait>.family.out`, and one for the unrelated subjects with name `<filename>.<trait>.unrelated.out`. Here, `<filename>` is the argument supplied to the **-tabularfile** option, and `<trait>` is the name of the trait analysed. The trait name is taken from the data file, but if no name is given then the default trait names are “Disease”

for a binary trait and “Trait” for a quantitative trait. The odds ratios (or additive values) in the tabular output are from the combined analysis of families and unrelated subjects, but the haplotype frequencies are for the families or unrelated subjects treated separately. To view the output files in HAPLOVIEW, select the “PLINK format” input with the “Non-SNP” option.

Additional options

The following options are not available from the GUI, and are primarily used for development. They are included here for completeness.

–checkgradient

Outputs analytic and numerical gradients of the log-likelihood evaluated at a random set of parameter values. Used for debugging the analytic gradient calculations.

–follow

Displays the log-likelihood after each evaluation in numerical maximisation.

–imputeDump

Outputs probabilities for individually imputed genotypes, given a test marker and tag markers. Currently implemented only for unrelateds with a single SNP as test marker. A dump file must be specified and this option overrides the usual output of the **–dump** option.

–llhd

Gives a complete dump of parameter values, log-likelihood contributions and gradients, for each subject and each likelihood evaluation. Used for debugging.

–neldermead

Uses Nelder & Mead’s downhill simplex method for likelihood maximisation, instead of the default Davidon-Fletcher-Powell. It is much slower, but can often get a better answer when DFP converges to the wrong solution.

–show

Displays the internal representation of haplotype solutions for all individuals, and the internal codes for haplotypes and genotypes. Mainly used for debugging.

–slow

Uses a slow algorithm for evaluating the denominator in the likelihood, fully enumerating all terms instead of factorising. Used for debugging. The **–genotype** and **–condgenotype** options automatically set this option.

Command line options

These are all the command line options with their minimal abbreviated forms and brief descriptions.

Option	Abbreviation	Description
-allcombinations	-allc	Test all combinations of a fixed number of markers
-allwindows	-allw	Test all window sizes
-baseline	-ba	Set baseline levels for factorial covariates
-bedfile	-be	Open a binary pedigree file
-brief	-br	Brief screen output
-cellcount	-cel	Identify rare haplotypes based on cell count
-certain	-cer	Only analyse certain haplotypes
-checkgradient	-che	Compare analytic and numerical gradients
-chrX	-chrX	Markers on chromosome X
-chrY	-chrY	Markers on chromosome Y
-compare	-com	First haplotype in comparison of two odds ratios
-condgenotype	-condg	Condition on genotypes
-condition	-condi	Select conditioning markers
-condspecific	-conds	Condition on a specific haplotype
-confounder	-conf	Select confounders
-covariance	-cov	Residual sibling covariance of quantitative trait
-datafile	-da	Open a data file
-disease	-di	Select affection status
-dumpfile	-du	Dump haplotypes to file
-epsilon	-e	Set convergence threshold
-factor	-fa	Designate covariates as factors
-follow	-fo	Display each evaluated log-likelihood
-genotype	-g	Analyse genotypes
-hhrr	-h	Unmatched family based controls
-imputeDump	-im	Output individually imputed genotypes
-individual	-in	Test individual haplotypes
-LD	-L	Show linkage disequilibrium estimates
-listmarkerfile	-li	Read markers to analyse from a file
-llhd	-lh	Output nearly all internal calculations
-mapfile	-map	Open a map file
-marker	-mar	Select test markers
-missing	-mi	Include uncertain haplotypes and missing genotypes
-model full	-mode full	Full haplotype coding model

-model gxg	-mode gxg	Test for gene-gene interaction
-model haplomain	-mode haplomain	Haplotype main effects model
-model allelemain	-mode allelemain	Allele main effects model
-model commonmain	-mode commonmain	Common main effect model
-modifier	-modi	Select effect modifiers
-mostlikely	-mos	Only dump the most likely haplotypes
-neldermead	-ne	Maximise likelihood with Nelder-Mead method
-nolinkage	-nol	Assume no linkage in nuclear families
-normal	-nor	Model explicit normal distribution for quantitative trait
-onefbc	-on	Use one family based control instead of three
-optionfile	-op	Open option file
-output	-ou	Save output to a file
-parentrisk	-pa	Model odds ratio in parents
-pedfile	-ped	Open pedigree file
-permoutput	-permo	Output permutation analyses
-permutation	-permu	Set number of permutations
-phenofile	-ph	Open a phenotype file
-quantile	-q	Show quantile from permutation distribution
-randomseed	-ran	Set random number seed
-rare	-rare	Set threshold for rare haplotypes
-reference	-ref	Reference haplotype
-restarts	-res	Number of random restarts in maximizing likelihood
-show	-sh	Show internal representation of possible haplotypes
-sibship	-si	Use a faster model for sibships
-slow	-sl	Slower calculation of denominator in likelihood
-specific	-sp	Test a specific haplotype
-tabularfile	-tab	Filename root for tabular output
-tag	-tag	Select tag markers
-testconfounders	-testc	Test confounder effects
-testmodifiers	-testm	Test modifier effects
-time	-ti	Output running time
-trait	-tr	Select quantitative trait
-uncentred	-un	Estimate quantitative trait mean from the data
-userare both	-us both	Define rare haplotypes from both cases & controls
-userare either	-us either	Define rare haplotypes from either cases or controls
-userare case	-us case	Define rare haplotypes from cases
-userare control	-us control	Define rare haplotypes from controls
-variance	-v	Quantitative trait variance
-window	-win	Sets number of markers for multi-locus analysis
-with	-wit	Second haplotype in comparison of odds ratio
-zero	-z	Set threshold for haplotypes with zero frequency

Methods

Likelihood

UNPHASED uses a retrospective likelihood, which is the probability of observing the parental and child genotypes, given the trait values of all the children in a nuclear family. This is appropriate when the data have been sampled based on the trait values of the children, which is usually true for case/parent trios, affected or discordant sib pairs, case/control samples and extreme sampling of quantitative traits. When other types of sampling occur, for example prospective sampling in a cohort study, the retrospective likelihood can still give valid results, but it is not the most efficient form of analysis.

In the following we will work with a quantitative trait Y , with specialization to a binary trait noted where appropriate. The retrospective likelihood contribution for a nuclear family with k children is

$$\Pr(f, m, \mathbf{c} | \mathbf{y}) = \frac{\Pr(f, m) \Pr(\mathbf{c} | f, m) \Pr(\mathbf{y} | f, m, \mathbf{c})}{\sum_{f^*, m^*, c_1^*, \dots, c_k^* \in G} \Pr(f^*, m^*) \Pr(\mathbf{c}^* | f^*, m^*) \Pr(\mathbf{y} | f^*, m^*, \mathbf{c}^*)} \quad (1)$$

where f and m are the genotypes of the father and mother, $\mathbf{c} = (c_1, \dots, c_k)^T$ is the vector of child genotypes, $\mathbf{y} = (y_1, \dots, y_k)^T$ is the vector of child traits, and G is the set of all genotypes. For the trait distribution we assume a multivariate normal model, with common mean μ and variance σ^2 for all children and residual covariance ρ between each pair of children. Denote the variance-covariance matrix by Σ . The retrospective nature of the likelihood gives some robustness to mis-specification of this model. Assuming random mating and Mendelian transmission we may write

$$\Pr(f, m, \mathbf{c} | \mathbf{y}) = \frac{\exp(\alpha_f + \alpha_m - \frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \beta[\mathbf{c}])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \beta[\mathbf{c}]))}{\sum_{f^*, m^*} \sum_{\mathbf{c} \in S(f^*, m^*)} \exp(\alpha_{f^*} + \alpha_{m^*} - \frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \beta[\mathbf{c}^*])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \beta[\mathbf{c}^*]))} \quad (2)$$

where α is a parameter vector for a multinomial distribution of genotypes, β is a parameter vector of additive genotype effects, $\beta[\mathbf{c}]$ denotes the vector $(\beta_{c_1}, \dots, \beta_{c_k})$, and $S(f^*, m^*)$ is the set of possible child genotype vectors from parents with genotypes f^* and m^* .

We prefer to work with the conditional likelihood of children given parents, because this is robust to population stratification. Writing

$$\frac{\exp(-\frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}]))}{\sum_{\mathbf{c}^* \in S(f, m)} \exp(-\frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*]))} \times \quad (3)$$

we have

$$\Pr(f, m, \mathbf{c} | \mathbf{y}) = \frac{\exp(-\frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}]))}{\sum_{\mathbf{c}^* \in S(f, m)} \exp(-\frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*]))} \times \quad (4)$$

$$\frac{\sum_{\mathbf{c}^* \in S(f, m)} \exp(\alpha_f + \alpha_m - \frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*]))}{\sum_{f^*, m^*} \sum_{\mathbf{c}^* \in S(f^*, m^*)} \exp(\alpha_{f^*} + \alpha_{m^*} - \frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*]))}$$

The innovation in UNPHASED is to model the genotype effect $\boldsymbol{\beta}$ separately in the two components of the likelihood. This means that we can perform inference based on the conditional likelihood while taking the parental distribution into account, which is necessary in order to deal with missing genotypes or uncertain haplotypes. We substitute new parameters $\tilde{\boldsymbol{\beta}}$ in place of $\boldsymbol{\beta}$ in the parental probability and write

$$\Pr(f, m, \mathbf{c} | \mathbf{y}) = \frac{\exp(-\frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}]))}{\sum_{\mathbf{c}^* \in S(f, m)} \exp(-\frac{1}{2}(\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}^*]))} \times \quad (5)$$

$$\frac{\sum_{\mathbf{c}^* \in S(f, m)} \exp(\alpha_f + \alpha_m - \frac{1}{2}(\mathbf{y} - \tilde{\mu} \mathbf{1} - \tilde{\boldsymbol{\beta}}[\mathbf{c}])' \Sigma^{-1} (\mathbf{y} - \tilde{\mu} \mathbf{1} - \tilde{\boldsymbol{\beta}}[\mathbf{c}]))}{\sum_{f^*, m^*} \sum_{\mathbf{c}^* \in S(f^*, m^*)} \exp(\alpha_{f^*} + \alpha_{m^*} - \frac{1}{2}(\mathbf{y} - \tilde{\mu} \mathbf{1} - \tilde{\boldsymbol{\beta}}[\mathbf{c}^*])' \Sigma^{-1} (\mathbf{y} - \tilde{\mu} \mathbf{1} - \tilde{\boldsymbol{\beta}}[\mathbf{c}^*]))}$$

Inference is now performed on $\boldsymbol{\beta}$, with $\tilde{\boldsymbol{\beta}}$ treated as nuisance parameters. In UNPHASED the default model sets $\tilde{\boldsymbol{\beta}} = \mathbf{0}$. This is appropriate for testing the null hypothesis that all genotypes have the same effects, but for other tests such as the conditional or gene-gene interaction tests, as well as for estimating $\boldsymbol{\beta}$, it is more accurate to freely estimate $\tilde{\boldsymbol{\beta}}$. It is also appropriate when testing association in the presence of linkage, as described below. Free estimation of $\tilde{\boldsymbol{\beta}}$ is specified by the **-parentrisk** option. The **-hhrr** option sets $\tilde{\boldsymbol{\beta}} = \boldsymbol{\beta}$, which results in the analysis being unconditional on parental genotypes. This is appropriate if there is no population stratification, especially if extra unrelated controls are available from the same population as the families.

Under the null hypothesis that $\boldsymbol{\beta} = \mathbf{0}$, the trait mean μ is not identifiable in the first (conditional) part of the likelihood, which leads to difficulties with the asymptotic theory; and when $\boldsymbol{\beta}$ is nonzero but small, μ is difficult to estimate numerically. To get around this problem, we write out the matrix product in the normal density as follows

$$\begin{aligned}
(\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}])' \Sigma^{-1} (\mathbf{y} - \mu \mathbf{1} - \boldsymbol{\beta}[\mathbf{c}]) &= \sum_{i,j} (y_i - \mu - \beta_{c_i}) s_{ij} (y_j - \mu - \beta_{c_j}) \\
&= \sum_{i,j} s_{ij} \left[(y_i - \mu)(y_j - \mu) + (y_i - \mu)\beta_{c_j} + (y_j - \mu)\beta_{c_i} + \beta_{c_i}\beta_{c_j} \right] \\
&= \sum_{i,j} s_{ij} \left[(y_i - \mu)(y_j - \mu) + y_i\beta_{c_j} + y_j\beta_{c_i} - \mu(\beta_{c_i} + \beta_{c_j}) + \beta_{c_i}\beta_{c_j} \right]
\end{aligned} \tag{6}$$

where s_{ij} is the i, j -th component of Σ^{-1} . In the retrospective likelihood, the first term in the bracket always cancels. Therefore the likelihood does not depend on μ when $\boldsymbol{\beta} = \mathbf{0}$. Now we add new nuisance parameters $\mathbf{v}$ to the other term involving μ , so we work with

$$\sum_{i,j} s_{ij} \left[y_i\beta_{c_j} + y_j\beta_{c_i} - \mu(\beta_{c_i} + \nu_{c_i} + \beta_{c_j} + \nu_{c_j}) + \beta_{c_i}\beta_{c_j} \right] \tag{7}$$

Here μ is still not identifiable as we can replace each $\mu(\beta_{c_i} + \nu_{c_i})$ with a single parameter, but those parameters are identifiable under all hypotheses about $\boldsymbol{\beta}$. When additional terms from covariates are present (see below), the above form (7) can be used explicitly, with μ and $\mathbf{v}$ identifiable and estimated.

In the parental part of the likelihood, we will not be testing $\tilde{\boldsymbol{\beta}} = \mathbf{0}$, so we can work with

$$\sum_{i,j} s_{ij} \left[y_i\tilde{\beta}_{c_j} + y_j\tilde{\beta}_{c_i} - \tilde{\mu}(\tilde{\beta}_{c_i} + \tilde{\beta}_{c_j}) + \tilde{\beta}_{c_i}\tilde{\beta}_{c_j} \right] \tag{8}$$

Here $\tilde{\mu}$ is assumed equal to 0 unless the **–uncentred** option is selected, in which case it is freely estimated.

UNPHASED automatically determines which parameters are identifiable and only estimates those that are identifiable.

The above model is the one implemented by UNPHASED under the **–normal** option. However, maximization of the likelihood can be time-consuming and numerically unstable, so this is not the default model. Some simplification is possible by assuming that the β 's are sufficiently small that second-order terms can be ignored. By further setting $\sigma=1$ and $\rho=0$, the default model of UNPHASED is

$$\Pr(f, m, \mathbf{c} | \mathbf{y}) = \frac{\exp(\mathbf{v}[\mathbf{c}] + \mathbf{y}'\boldsymbol{\beta}[\mathbf{c}])}{\sum_{\mathbf{c}^* \in S(f, m)} \exp(\mathbf{v}[\mathbf{c}^*] + \mathbf{y}'\boldsymbol{\beta}[\mathbf{c}^*])} \cdot \frac{\sum_{\mathbf{c}^* \in S(f, m)} \exp(\alpha_f + \alpha_m)}{\sum_{f^*, m^*} \sum_{\mathbf{c}^* \in S(f^*, m^*)} \exp(\alpha_{f^*} + \alpha_{m^*})} \tag{9}$$

For binary traits, it is convenient to code $y_i = 1$ for affected, 0 for unaffected. Setting $\mu = 0$ gives the same conditional likelihood as used in standard models of the TDT, and β may be interpreted as log odds ratios.

When data are missing, we form the above probabilities for each possible completion and sum the probabilities to give the likelihood contribution for the family.

For analysis of sibships, the same model is used except that the parental genotype frequencies are approximated by the average of the sibling genotype frequencies. That is, where $\alpha_f + \alpha_m$ appear in the likelihood, we now use $\frac{2}{k} \mathbf{1} \cdot \alpha[\mathbf{c}]$. Furthermore, we also condition on the sets of haplotypes transmitted into the sibship by each parent, meaning that when a parent cannot be deduced from the sibs, the likelihood contribution is the same for all completions of the missing parent. This option speeds up the analysis of sibships with two missing parents, with a small cost in power. Untransmitted haplotypes do not enter into the likelihood, and the sibship is only informative if it contains more than one genotype and more than one phenotype. This method is described in detail in the paper by Dudbridge et al, *Ann Hum Genet* 75:428-38

For unrelated subjects, we assume two missing parents and set $\tilde{\mathbf{v}} = \mathbf{v}$ and $\tilde{\beta} = \beta$, giving likelihood contribution

$$\Pr(c | y) = \frac{\exp(\alpha_c - \frac{1}{2} \sigma^{-2} (y\beta_c - \mu(\beta_c + v_c) + \beta_c^2))}{\sum_{c^* \in G} \exp(\alpha_{c^*} - \frac{1}{2} \sigma^{-2} (y\beta_{c^*} - \mu(\beta_{c^*} + v_{c^*}) + \beta_{c^*}^2))} \quad (10)$$

Here again μ is only identifiable if extra covariate terms are included, and in fact $\mathbf{v}$ is confounded with α and is not identifiable itself unless the unrelated subjects are jointly analysed with a family sample.

This may also be simplified by omitting the second-order term and setting $\sigma=1$. For a binary trait this then becomes a standard retrospective likelihood for case-control data

$$\Pr(c | y) = \frac{\exp(\alpha_c + y\beta_c)}{\sum_{c^* \in G} \exp(\alpha_{c^*} + y\beta_{c^*})} \quad (11)$$

in which β may again be interpreted as log odds ratios and α are multinomial logistic parameters for the genotype distribution in controls.

So far we have assumed that transmissions to the children are independent. This actually corresponds to the **-nolinkage** option in UNPHASED, but the default behaviour allows for dependence amongst transmissions, such as when there is prior linkage. This is done by conditioning on the inheritance vector of the observed data, in the conditional part of

the likelihood. For each child c_i of parents f and m , construct three *virtual genotypes* as follows:

u_i^m : haplotype transmitted by father + haplotype not transmitted by mother

u_i^f : haplotype not transmitted by father + haplotype transmitted by mother

u_i^{fm} : haplotype not transmitted by father + haplotype not transmitted by mother

Then the vectors of child genotypes $\mathbf{u}^m$, $\mathbf{u}^f$ and $\mathbf{u}^{fm}$ have the same inheritance vector as the observed family, up to an equivalence class. Conditioning on this inheritance vector changes the summations over possible sibships. The normal likelihood becomes

$$\Pr(f, m, \mathbf{c} | \mathbf{y}) = \frac{\exp(\mathbf{v}[\mathbf{c}] - \frac{1}{2}(\mathbf{y} - \boldsymbol{\beta}[\mathbf{c}])' \Sigma^{-1}(\mathbf{y} - \boldsymbol{\beta}[\mathbf{c}]))}{\sum_{\mathbf{c}^* \in \{\mathbf{c}, \mathbf{u}^m, \mathbf{u}^f, \mathbf{u}^{fm}\}} \exp(\mathbf{v}[\mathbf{c}^*] - \frac{1}{2}(\mathbf{y} - \boldsymbol{\beta}[\mathbf{c}^*])' \Sigma^{-1}(\mathbf{y} - \boldsymbol{\beta}[\mathbf{c}^*]))} \times \frac{\sum_{\mathbf{c}^* \in \{\mathbf{c}, \mathbf{u}^m, \mathbf{u}^f, \mathbf{u}^{fm}\}} \exp(\alpha_f + \alpha_m + \tilde{\mathbf{v}}[\mathbf{c}^*] - \frac{1}{2}(\mathbf{y} - \boldsymbol{\beta}[\mathbf{c}^*])' \Sigma^{-1}(\mathbf{y} - \boldsymbol{\beta}[\mathbf{c}^*]))}{\sum_{f^*, m^*} \sum_{\mathbf{c}^* \in S(f^*, m^*)} \exp(\alpha_{f^*} + \alpha_{m^*} + \tilde{\mathbf{v}}[\mathbf{c}^*] - \frac{1}{2}(\mathbf{y} - \boldsymbol{\beta}[\mathbf{c}^*])' \Sigma^{-1}(\mathbf{y} - \boldsymbol{\beta}[\mathbf{c}^*]))} \quad (12)$$

with the same simplifications as above for small effects and binary traits. Note however that if $\tilde{\boldsymbol{\beta}} = \mathbf{0}$, the parental part of the likelihood is proportional to that obtained when not conditioning on the inheritance vector, so the **-parentrisk** option should be used to maintain robustness. This approach has the effect of treating the whole family as one sampling unit, in which unmodelled correlation is absorbed into the error variance.

Covariates are included by adding terms to the linear model. UNPHASED only includes first-order terms for covariates. For a factorial confounder at level z_i in child i we add a parameter $\gamma_{g;z_i}$ to terms involving α_g , and for a continuous confounder with value z_i we add a term $z_i \gamma_g$. The γ parameters reflect changes in haplotype frequency and population mean according to different covariate values, though they may not always be interpreted directly as such. For a factorial effect modifier at level z_i we add a parameter $y_i \eta_{g;z_i}$ to terms involving β_g , and for a continuous modifier with value z_i we add a term $y_i z_i \eta_g$. Here the η parameters reflect changes in odds ratio or additive value, on a linear scale with trait value. These are the odds ratios or additive values displayed in the output.

UNPHASED conditions on all the covariate values, so there is no “main effect” of a covariate. This means that all analyses are automatically adjusted for average effects of covariates. This is often more conditioning than we should need, but it results in simpler computations, particularly for continuous covariates.

To summarise, the default model assumes small genotype effects $\boldsymbol{\beta}$, unit variance ($\sigma^2 = 1$), no residual covariance ($\rho = 0$), zero-centred traits ($\tilde{\mu} = 0$), no genotype effects

in the parental likelihood ($\tilde{\beta} = \mathbf{0}$), and conditions on the inheritance vector. To use more complex likelihoods in UNPHASED, the **–normal**, **–variance**, **–covariance**, **–uncentred**, **–parentrisk** and **–nolinkage** options may be applied to relax these respective restrictions. The **–normal** and **–parentrisk** options both increase the computation time. The **–normal** option may encounter problems of numerical convergence, which can be addressed with the **–restarts** and **–neldermead** options, both of which increase the running time again.

The variance is left as an input parameter, rather than being estimated by UNPHASED. This is because numerical maximization of the variance is somewhat unreliable, basically because the mean is not always identifiable. Also in the situation of $\beta = \mathbf{0}$, the variance itself is not identifiable, leading to problems with the asymptotic theory. The variance could however be estimated by profiling the likelihood across a range of values, to give a rough idea of its true value. Similar comments apply to the residual covariance.

The individual score tests are calculated from the partial derivatives of the log-likelihood, evaluated at the null hypothesis that $\beta = \mathbf{0}$. For simplicity assume that no additional

covariates are included. Let the score vector be $\mathbf{U} = \left(\frac{\partial \log L}{\partial \beta_g}, \frac{\partial \log L}{\partial \alpha_g} \right)'$ with variance-

covariance matrix $\mathbf{V} = \left(\frac{\partial \log \mathbf{U}}{\partial \beta_g}, \frac{\partial \log \mathbf{U}}{\partial \alpha_g} \right)^{-1}$. Let $\mathbf{U}_\beta$ be the sub-vector of $\mathbf{U}$

corresponding to the parameters of interest β , with $\mathbf{V}_{\beta\beta}$ be the corresponding sub-matrix of $\mathbf{V}$. Let $\mathbf{V}_{\beta\alpha}$, $\mathbf{V}_{\alpha\alpha}$ and $\mathbf{V}_{\alpha\beta}$ be the appropriate sub-matrices of $\mathbf{V}$ corresponding to the nuisance parameters α . Then the variance-covariance matrix for $\mathbf{U}_\beta$ is

$\mathbf{V}_{\beta\beta} - \mathbf{V}_{\beta\alpha}(\mathbf{V}_{\alpha\alpha})^{-1}\mathbf{V}_{\alpha\beta}$ and the χ^2 score statistic for genotype g is $(\mathbf{U}_\beta)_g^2 (\mathbf{V}_{\beta\beta})_{gg}^{-1}$.

When evaluated at maximum-likelihood estimates of β , the above expression for $\mathbf{V}_\beta$ is used to obtain 95% confidence intervals according to $\beta_g \pm 1.96(\mathbf{V}_{\beta\beta})_{gg}^{1/2}$. Note that as of version 3.0.12, $\mathbf{V}$ is calculated from the Fisher information rather than the previously used outer-product estimator. This should give more accurate confidence intervals and agree with those obtained from other software packages.

Haplotype coding

The above description is in terms of genotypes, for clarity. In fact UNPHASED deals primarily with haplotypes, although genotype tests are also implemented. Basically this replaces each genotype parameter by a sum of two haplotype parameters. This imposes some assumptions of independence between haplotypes: Hardy-Weinberg equilibrium, random mating, multiplicative odds ratio, and independent risk between siblings.

However, owing to the likelihood factorisation above, inference on β is relatively unaffected when these assumptions are violated.

There are two paradigms for parameterising the haplotype odds ratios. *Haplotype coding* simply assigns an odds ratio parameter to each observed haplotype. *Allele coding* represents haplotype risks in terms of main effects of each locus and interactions between them. Haplotype coding is more intuitive to geneticists, but allele coding is more natural to statisticians. UNPHASED uses either version when convenient.

Consider three SNPs with eight haplotypes. The parameterisations under the two schemes could be represented as follows:

Haplotype	Haplotype coding	Allele coding
1-1-1	β_{111}	β_0
1-1-2	β_{112}	$\beta_0 + \beta_{-2}$
1-2-1	β_{121}	$\beta_0 + \beta_{-2-}$
1-2-2	β_{122}	$\beta_0 + \beta_{-2-} + \beta_{-2} + \beta_{-22}$
2-1-1	β_{211}	$\beta_0 + \beta_{2--}$
2-1-2	β_{212}	$\beta_0 + \beta_{2--} + \beta_{-2} + \beta_{2-2}$
2-2-1	β_{221}	$\beta_0 + \beta_{2-} + \beta_{2-} + \beta_{22-}$
2-2-2	β_{222}	$\beta_0 + \beta_{2--} + \beta_{-2-} + \beta_{-2} + \beta_{22-} + \beta_{2-2} + \beta_{-22} + \beta_{222}$

Both schemes involve eight parameters. A baseline parameter must be set to 0 to ensure identifiability. Haplotype coding is the default in UNPHASED, and is called the **full** model. By dropping the interaction terms, allele coding allows models to be fitted with fewer parameters, which is particularly useful for the conditioning tests. The **allelemain** model in UNPHASED would code the eight haplotypes with just four parameters, a baseline and three main effects:

Haplotype	Allele main effects
1-1-1	β_0
1-1-2	$\beta_0 + \beta_{-2}$
1-2-1	$\beta_0 + \beta_{-2-}$
1-2-2	$\beta_0 + \beta_{-2-} + \beta_{-2}$
2-1-1	$\beta_0 + \beta_{2--}$
2-1-2	$\beta_0 + \beta_{2--} + \beta_{-2}$
2-2-1	$\beta_0 + \beta_{2-} + \beta_{2-}$
2-2-2	$\beta_0 + \beta_{2-} + \beta_{-2-} + \beta_{-2}$

This can lead to a global test with fewer degrees of freedom than the full model. When this model is fitted, UNPHASED displays the haplotype odds ratios estimated by the above model, and does not display the actual parameters estimated.

The **commonmain** model goes further and constrains the allele main effects to be equal, giving just two parameters, a baseline and one main effect:

Haplotype	Common main effects
1-1-1	β_0
1-1-2	$\beta_0 + \beta_2$
1-2-1	$\beta_0 + \beta_2$
1-2-2	$\beta_0 + 2\beta_2$
2-1-1	$\beta_0 + \beta_2$
2-1-2	$\beta_0 + 2\beta_2$
2-2-1	$\beta_0 + 2\beta_2$
2-2-2	$\beta_0 + 3\beta_2$

When there are conditioning markers as well as test markers, UNPHASED can mix the two coding schemes. The **haplomain** model defines a full model for the conditioning markers, and another full model for the test markers, but has no interaction terms between the two sets of markers. Considering now two conditioning SNPs and two test SNPs, the 16 haplotypes are coded as follows:

Haplotype	Haplotype main effects
1-1-1-1	β_0
1-1-1-2	$\beta_0 + \beta_{-12}$
1-1-2-1	$\beta_0 + \beta_{-21}$
1-1-2-2	$\beta_0 + \beta_{-22}$
1-2-1-1	$\beta_0 + \beta_{12--}$
1-2-1-2	$\beta_0 + \beta_{12--} + \beta_{--12}$
1-2-2-1	$\beta_0 + \beta_{12--} + \beta_{--21}$
1-2-2-2	$\beta_0 + \beta_{12--} + \beta_{--22}$
2-1-1-1	$\beta_0 + \beta_{21--}$
2-1-1-2	$\beta_0 + \beta_{21--} + \beta_{--12}$
2-1-2-1	$\beta_0 + \beta_{21--} + \beta_{--21}$
2-1-2-2	$\beta_0 + \beta_{21--} + \beta_{--22}$
2-2-1-1	$\beta_0 + \beta_{22--}$
2-2-1-2	$\beta_0 + \beta_{22--} + \beta_{--12}$
2-2-2-1	$\beta_0 + \beta_{22--} + \beta_{--21}$
2-2-2-2	$\beta_0 + \beta_{22--} + \beta_{--22}$

So this coding has 7 parameters instead of 16 in the full model. In this case the **haplomain** test would have 3df corresponding to the free parameters for the test haplotype. If the **allelemain** model is used, there would be 2df, reflecting the allele coding for the test markers. If the **full** model is used, there would be 3df for each of the four conditioning haplotypes, resulting in 12df in total. The **gxx** test compares the haplotype main effects model to the full model. Here the test would have 9df.

Frequently asked questions

I have unaffected as well as affected offspring. How are the untransmitted counts calculated?

(See pages 14-15) The counts include all the untransmitted haplotypes and the haplotypes transmitted to unaffecteds. The transmitted counts only include the haplotypes transmitted to affecteds.

The transmitted and untransmitted counts add up to different numbers. How do I get transmitted/untransmitted counts for a TDT?

(See pages 14-15) The counts include homozygous parents, and trans/untrans counts for TDT are unfortunately not available. This is because when a parent is missing, we don't know whether or not it is homozygous. One workaround is to assume Hardy-Weinberg equilibrium and subtract the expected homozygous counts from the case and control counts.

How are unrelateds combined with trios?

(See page 41) A single model is fit for the haplotype frequencies and odds ratios. The likelihood contributions are different for the different data types, but can be written in a unified form with the same parameters appearing in the likelihoods for different data types. You'll have to read the paper to get this exactly.

I have trios together with extra unaffected controls. How are they combined?

In the default analysis, the extra controls only improve the resolution of missing data. If you are doing single locus analysis in complete trios, extra controls make no difference. This is because the extra controls give no information on within-family association. However the **-hhrr** option can be used to break the matching within families, and extra controls then give information on association. This analysis is less robust to population stratification than the default, and is not robust to prior linkage.

What threshold should I use for rare haplotypes?

There is no recommendation with solid theory, but many users use 5%.

Should I use the **-parentrisk option?**

The option is useful when testing multiple affected sibs in the presence of linkage, and there are missing parents. It is also useful for conditional tests and for estimating odds ratios, again when parents are missing.

How do I test under dominant/recessive models?

These models are hard to define when there are more than 2 alleles/haplotypes. Combining the `--genotype` and `--individual` options can give recessive tests for each allele.

Shouldn't `--chrX` treat males as homozygotes?

This only makes sense when there are 2 alleles. Also we want the reported odds ratio to mean the same thing when the data are all males or all females, which is best achieved by the current coding.

The individual test is significant, but the confidence interval overlaps 1.

(See page 16) The individual test compares the haplotype to all others combined, whereas the confidence interval compares to the reference haplotype. Also, the individual test is a score test whereas the confidence interval relates to a Wald test. To get a confidence interval relative to all others combined, use the `--specific` option.

The confidence interval excludes 1, but the individual test is not significant.

See above. To test a haplotype relative to the reference, using the `--compare` and `--with` options.

The results are different from version 2.4.

The family-based analysis is fundamentally different in version 3.0. In analysis of unrelates, version 2.4 had an unresolved bug in handling missing genotypes, which nevertheless did not affect type-1 error.

I used to use PDTPHASE. How can I use UNPHASED 3 in general pedigrees?

The PDT is not likelihood based so is not included in UNPHASED 3. However the current version can test for association in the presence of linkage when parents are missing, so can often be used instead of the PDT. PDTPHASE is still supported at version 2.4.

I don't understand the output for covariates.

(See pages 15 and 44) The reported coefficients are relative to the baseline level. For an effect modifier the linear model used has the form $y(\beta_g + \eta_{g,z})$ where g is the genotype and z is the covariate level. "Odds-R" is η and reflects a change in effect size for modifiers, relative to the reference level at which the effect size is β_g .

How many permutations should I run? How long is it going to take?

At least 1000 for accurate computation, ideally 10,000. But start with 1 or 10 permutations to get an idea of the running time.

I am also using FBAT/TRANSMIT/PDT/PLINK etc. Sometimes UNPHASED is more significant and sometimes less. Which should I believe?

We expect one program to be more significant about half of the time. If there is a discrepancy of orders of magnitude in the P-values, but nearly all other aspects of the analysis are equal (particularly the amount of missing data), then feel free to send me your input and output files.

Is there a parallel version of UNPHASED?

Not yet. The main application for a parallel version would be permutation tests, for which I usually start several short runs in parallel with different random number seeds, and collate the results at the end.

Is there a Macintosh version of UNPHASED?

I don't have one, but if you compile UNPHASED for Macintosh, feel free to email me the binary and I'll distribute it.

Can I analyse a genome-wide association scan with UNPHASED?

From version 3.1.0, UNPHASED can store whole chromosome files in memory and can read binary pedigree files created by PLINK. Note that the supplied Windows binary cannot handle large pedigree files (if you manage to get a working version, please send me a copy). Fast analysis of SNPs can be achieved using the options **-zero 0 -individual -model null**. However, UNPHASED is not designed as a GWAS package. In particular, none of the usual quality control filters are implemented. A number of excellent packages for GWAS analysis are available, including PLINK, GENABEL, SNPASSOC and SNPSTATS. I suggest that these packages are used for the initial quality control and analysis, and UNPHASED used to verify promising results in situations where its methods are more powerful, such as haplotype analysis and family-based studies with missing parents.

Differences from version 2.4

This version of UNPHASED is almost a complete rewrite of the previous version, 2.4. In addition to a number of new options, the major differences are as follows:

- Merging of TDTPHASE, COCAPHASE and QTPHASE into a single program that can analyse combined samples of trios and unrelateds.
- New likelihood used in TDTPHASE that is more robust to deviation from HWE. It reduces to the TDT when data are complete, whereas the old TDTPHASE reduced to the HHRR when the `-EM` option was used.
- Robustness to linkage in nuclear families with multiple siblings. Previously this was only available with the `-robustperm` option.
- Outputs confidence intervals for the odds ratio.
- Genotype tests now implemented.
- Basic support for covariates.
- Score tests for `-individual` option, instead of likelihood ratio test.
- Automated analysis of all combinations of a fixed number of markers, and automated analysis of all window sizes.
- QTDT format data file.
- Test for gene-gene interaction.
- New graphical user interface.
- `-EM` dropped and `-certain` introduced instead.
- `-bygenotype` renamed to `-condgenotype`.
- `-cellcounts` renamed to `-cellcount`.
- `-conditiontype` renamed to `-condspecific`.
- `-hpt` dropped.
- `-maineffects` and `-nointeraction` renamed to `-model haplomain` and `-model allelemain`.
- `-observed` dropped.
- `-onesib` dropped.
- `-paraff` and `-sibaff` dropped.
- `-parsex` and `-sibsex` dropped and now used as covariate names.
- `-phenotype` renamed to `-trait`.
- `-robustperm` dropped.

The PDTPHASE and QPDTPHASE programs from version 2.4 are still supported and available in that version. Because their methods are not likelihood based, they have not been incorporated into version 3.